Physics-Guided Latent Diffusion for Lensless Depth from PSF Stacks

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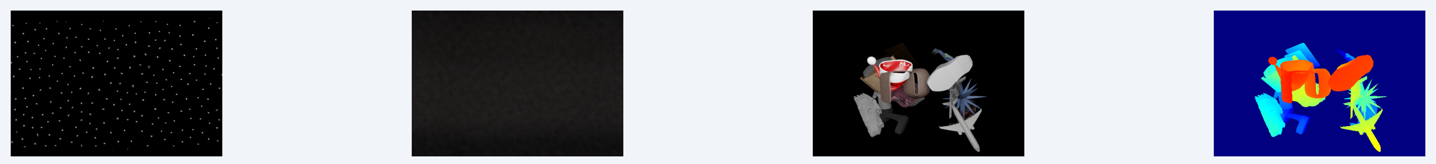
- A lensless measurement mixes RGB radiance through depth-dependent PSFs.
- We estimate depth from one raw measurement and a calibrated 42-plane PSF stack.
- The central cue is focus variation across PSF-Wiener deconvolution planes.

Data and Forward Physics

Training samples contain RGB ground truth, normalized depth labels, and a calibrated 42-plane PSF stack. Lensless raw inputs are synthesized by depth-sliced convolution:

$$y = \sum_{k=1}^{42} (w_k(d) \odot x) * h_k + \eta,$$

where $w_k(d)$ softly bins each pixel depth into neighboring PSF planes.



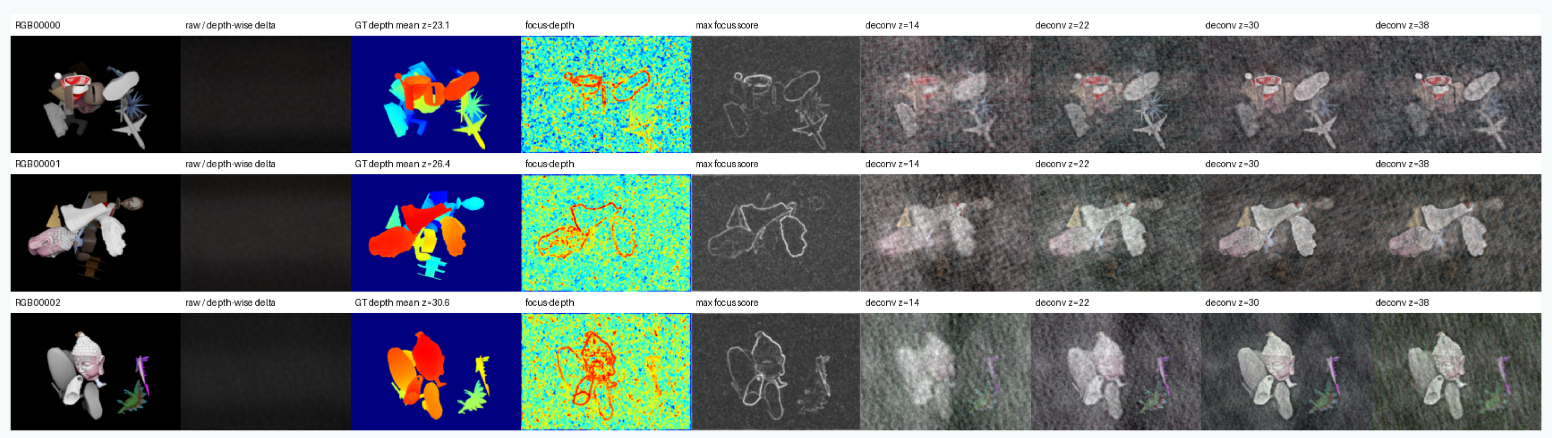
PSF plane lensless raw GT RGB GT depth

PSF-Stack Deconvolution

For each depth plane, we compute a Wiener inverse:

$$\hat{x}_k = \mathcal{F}^{-1}\left(\frac{\overline{H_k}Y}{|H_k|^2 + \lambda_k}\right).$$

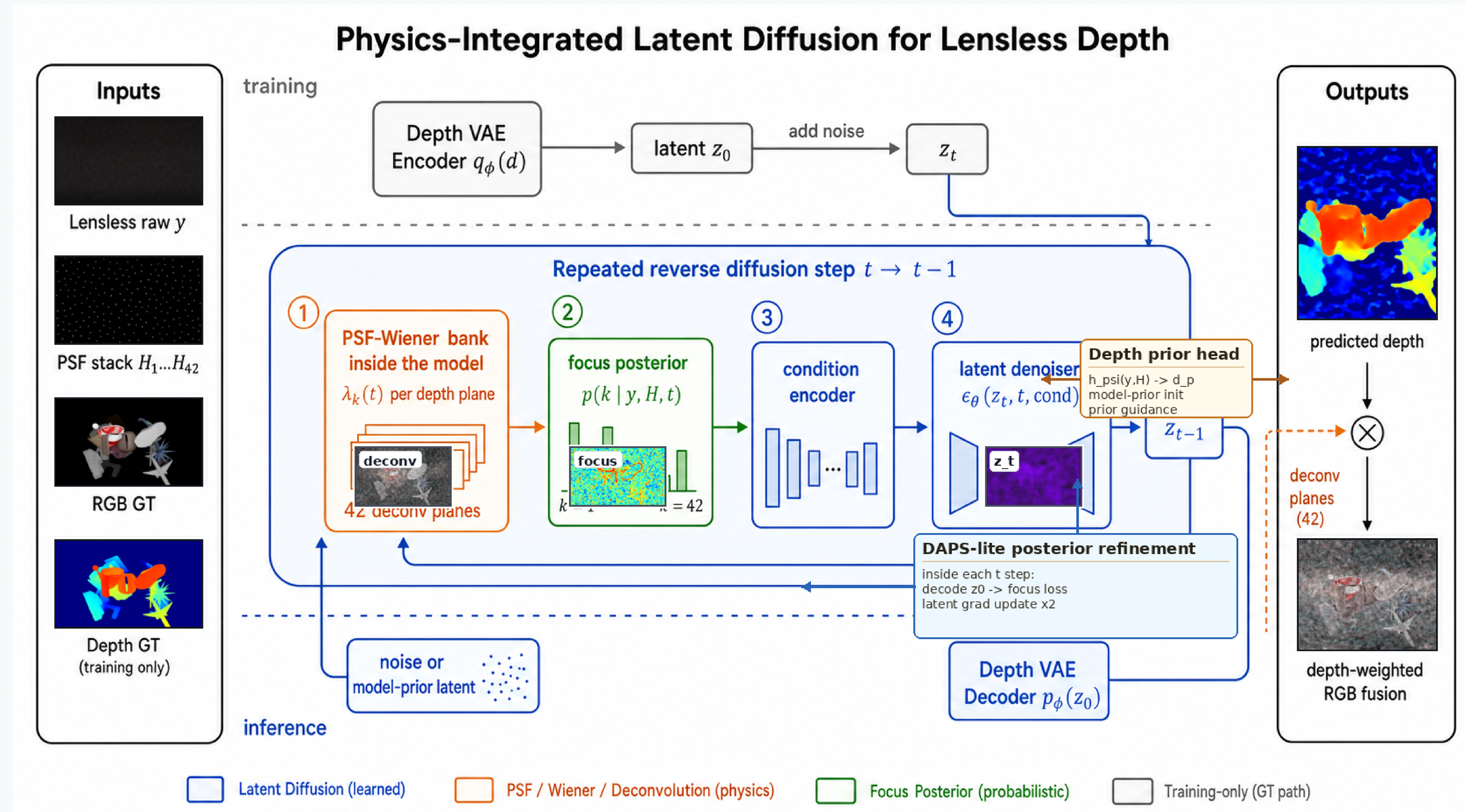
Early depth planes produce weak or unstable inverses, so the poster visualizes only later useful planes. The four panels below use $z = \{14, 22, 30, 38\}$.



Mid/late deconvolution planes reveal object contours and depth-dependent focus without unstable early-plane inverses.

Ours is v15 DAPS-lite posterior-refinement latent diffusion. Each reverse step:

1. predicts a clean depth latent $\hat{z}_0$ from noisy z_t;
2. decodes $\hat{z}_0$ to metric depth;
3. evaluates focus likelihood from the PSF-Wiener deconvolution stack;
4. backpropagates the focus residual into z_t;
5. repeats two short posterior refinements, then continues the DDIM transition.



Latent diffusion sampler with PSF-Wiener focus likelihood. Real project PSF/raw/deconvolution/focus/RGB/depth/output panels are embedded.

Why Diffusion?

Diffusion treats depth as a latent variable sampled under two forces:

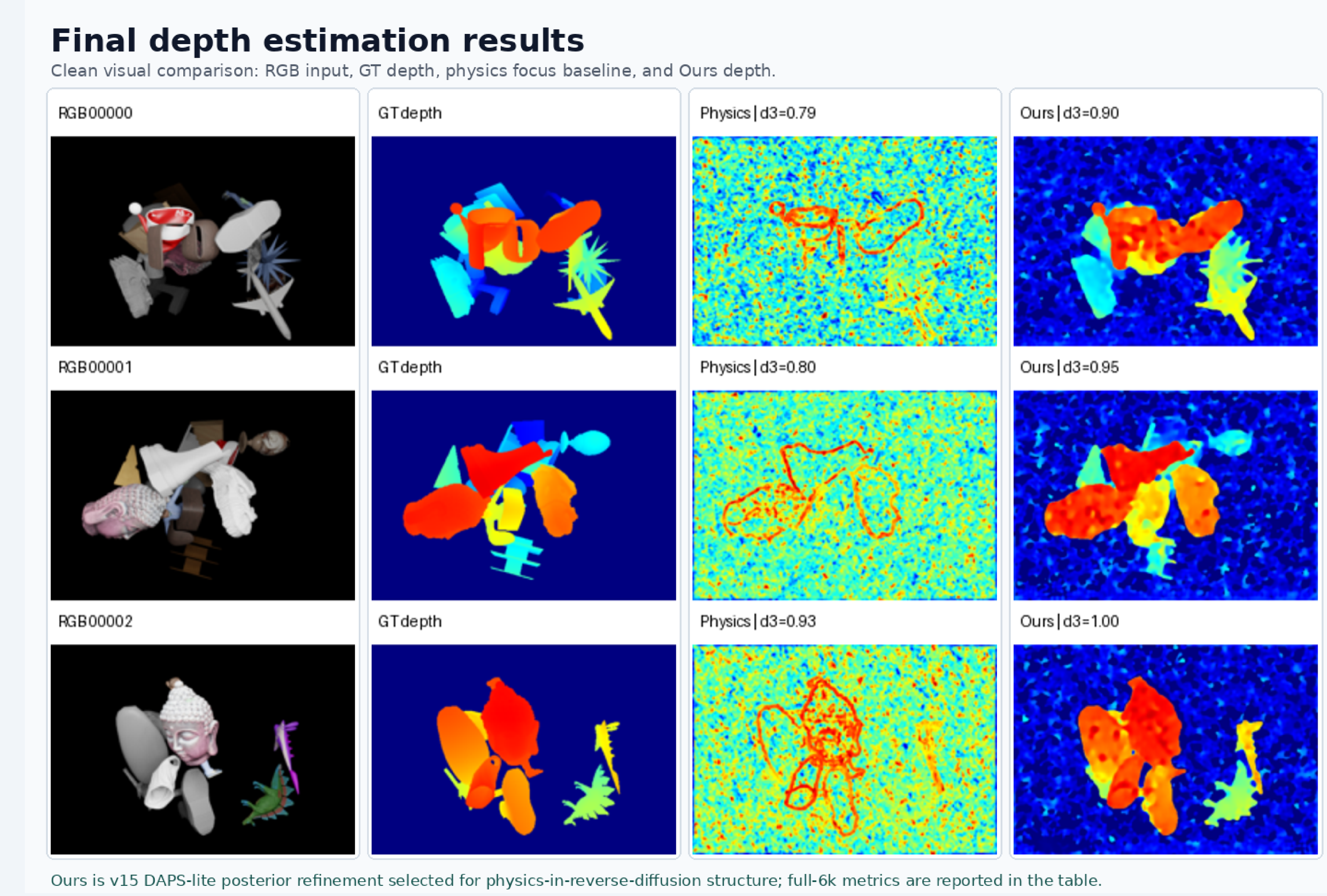
$$\nabla_{z_t} \log p(z_t \mid y) \approx \nabla_{z_t} \log p_\theta(z_t) - \alpha_t \nabla_{z_t} \mathcal{L}_{\text{focus}}(D(\hat{z}_0), y).$$

The learned denoiser supplies a depth prior; the PSF-Wiener focus term pulls samples toward depths that explain the lensless measurement. RGB restoration is then produced by depth-weighted deconvolution-plane fusion.

We evaluate foreground depth metrics on the 6,000-sample test set. The final **Ours** row is selected by method structure: physics is applied inside the latent reverse process.

Method	Eval	δ_1	δ_3	MAE
Physics focus	6k	0.391	0.816	0.214
Raw U-Net, 66k	6k	0.436	0.753	0.204
Deconv U-Net, 66k	6k	0.865	0.949	0.050
Deconv U-Net + affine	6k	0.881	0.968	0.046
Earlier integrated LDM v5	6k	0.878	0.948	0.059
Ours: v15 DAPS-lite	6k	**0.871**	**0.930**	**0.057**

Qualitative Results



Final visual panel: RGB, GT depth, physics focus baseline, and Ours depth only.

Findings

- Raw input alone is weak; PSF-Wiener deconvolution is the useful depth representation.
- Ours is less metric-dominant than direct supervised regression, but it is the cleanest physics-in-reverse-diffusion model.
- The remaining gap is mainly calibration and boundary ambiguity, not total depth failure.

Takeaway

PSF-stack deconvolution is the right representation for lensless depth. **Ours** demonstrates how to move the same optical physics inside a latent diffusion posterior sampler. Future work should tune sigma-aware clean-latent correction and plane-to-depth calibration.

Final Ours metric v15 full-6k: $\delta_1 = 0.871$, $\delta_3 = 0.930$, MAE $= 0.057$.